

Gunshot Detection and Direction Estimation Using UAV

Project Report

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Abstract

Gun-related incidents and illegal firearm activities pose significant threats to public safety, wildlife protection, and military surveillance systems. Traditional surveillance techniques based on optical cameras suffer from poor performance under low-light, foggy, smoky, or occluded environments. This work presents a UAV-based acoustic gunshot detection and direction estimation system using signal processing and machine learning techniques. The proposed system employs a four-microphone array mounted on a quadcopter platform to capture impulsive acoustic events. Time Difference of Arrival (TDOA), onset timing, spectral features, and intensity ratios are extracted from multi-channel recordings. A Random Forest classifier is trained to estimate the direction of the gunshot source across multiple angular classes. Additionally, a gunshot detection pipeline using VGGish embeddings and neural network classification is implemented to distinguish gunshots from other impulsive sounds. Experimental evaluation demonstrates a gunshot detection accuracy of 92.47% and a direction estimation accuracy of 84.6% across 12 directional classes. The proposed system shows the feasibility of real-time UAV-based acoustic surveillance for security and emergency-response applications.

0.1 Introduction

Gunshot detection and localization systems have gained significant importance in modern surveillance and security applications. Rapid identification of firearm discharge can improve emergency response time, support law enforcement operations, and assist in wildlife anti-poaching missions. Traditional surveillance systems primarily depend on cameras and fixed infrastructure. However, optical systems exhibit several limitations in low-visibility conditions such as darkness, smoke, fog, and cluttered urban environments. Acoustic sensing offers an alternative approach because sound propagation is independent of lighting conditions. Recent advancements in Unmanned Aerial Vehicles (UAVs) have enabled the deployment of mobile surveillance systems capable of covering large geographic areas. UAV-mounted acoustic systems provide mobility, flexibility, and rapid deployment capability in inaccessible terrains. This work proposes a UAV-based gunshot detection and direction estimation framework using a four-microphone array mounted on a quadcopter platform. Signal processing techniques including cross-correlation-based Time Difference of Arrival (TDOA), onset detection, and spectral analysis are combined with machine learning models for robust gunshot localization.

The primary contributions of this work are:

- Design of a UAV-mounted four-microphone acoustic array
- TDOA-based direction estimation framework
- Random Forest-based multi-class direction classification
- Real-time gunshot detection pipeline
- Experimental validation using custom multi-channel recordings

0.2 Problem Statement

Despite significant advancements in surveillance technology, reliable gunshot detection and localization remain challenging problems due to several environmental and operational factors. Urban noise, environmental interference, reverberation, wind noise, and drone propeller noise can significantly affect acoustic recordings. In addition, accurate localization of impulsive sound sources requires precise synchronization and robust spatial feature extraction techniques. Most existing systems rely on fixed ground-based sensor networks that provide limited spatial coverage and reduced operational flexibility. Furthermore, many conventional systems require expensive hardware infrastructure

and complex deployment procedures. Therefore, there is a need for a portable, low-cost, real-time UAV-based acoustic surveillance system capable of:

- Detecting gunshot events reliably
- Distinguishing gunshots from other impulsive sounds
- Estimating the direction of the sound source
- Operating effectively under low-visibility conditions
- Supporting real-time onboard processing using Jetson Nano

0.3 Motivation

The primary motivation behind this work is to develop a mobile acoustic surveillance framework capable of autonomous gunshot detection and localization using UAV technology. Acoustic localization systems mounted on UAVs offer several advantages over traditional fixed surveillance infrastructure. First, UAV-based systems provide dynamic mobility and wider area coverage, allowing surveillance operations in remote or inaccessible regions. Second, UAV-mounted systems can rapidly respond to suspicious acoustic events and provide real-time situational awareness. Third, acoustic sensing remains functional under visually degraded conditions where optical systems fail. Another important motivation is the growing availability of embedded AI hardware platforms such as the NVIDIA Jetson Nano, which enable onboard real-time machine learning inference for autonomous surveillance applications. This work also aims to investigate the effectiveness of combining classical signal processing techniques with machine learning methods for impulsive sound localization problems.

0.4 Objectives

The major objectives of this research work are as follows:

1. Develop a UAV-mounted multi-microphone acoustic sensing platform
2. Detect gunshot events from environmental audio recordings
3. Distinguish gunshots from acoustically similar impulsive sounds
4. Extract spatial acoustic features using TDOA analysis
5. Estimate gunshot direction using machine learning techniques

6. Implement a real-time onboard processing framework
7. Evaluate the performance of the proposed system experimentally

0.5 Major Contributions

The primary contributions of this work are summarized below:

- Design and implementation of a four-microphone UAV acoustic array
- Development of a TDOA-based spatial localization framework
- Integration of VGGish-based audio feature extraction for gunshot classification
- Development of a Random Forest-based direction estimation model
- Creation of a custom multi-channel directional audio dataset
- Implementation of a real-time live gunshot direction detection pipeline
- Experimental validation of acoustic localization performance

Chapter 1

Literature Review

1.1 Overview of Acoustic Gunshot Detection

Acoustic gunshot detection systems have been widely studied for military surveillance, urban security, border monitoring, and wildlife protection applications. The primary objective of these systems is to detect impulsive firearm discharge events and estimate the location or direction of the source using acoustic sensing techniques. Early gunshot detection systems primarily relied on threshold-based signal processing methods using peak amplitude analysis, energy detection, and spectral characteristics. Although these approaches were computationally simple, they suffered from high false-positive rates in noisy environments. Modern systems combine advanced signal processing techniques with machine learning algorithms to improve robustness and classification accuracy. Feature extraction methods such as Mel-Frequency Cepstral Coefficients (MFCC), spectrogram analysis, wavelet transforms, and deep neural network embeddings are widely used for acoustic event classification.

1.2 Time Difference of Arrival (TDOA) Localization

Time Difference of Arrival (TDOA) is one of the most widely used techniques for acoustic source localization. TDOA-based systems estimate the relative propagation delay between microphone pairs to infer the direction or position of the sound source. For two microphone signals $x(t)$ and $y(t)$, the cross-correlation function is expressed as:

$$R_{xy}(\tau) = \int_{-\infty}^{\infty} x(t) y(t + \tau) dt$$

For discrete-time sampled signals, the cross-correlation is written as:

$$R_{xy}[k] = \sum_{n=-\infty}^{\infty} x[n] y[n+k]$$

The estimated delay corresponds to the lag value that maximizes the cross-correlation magnitude:

$$\tau^* = \arg \max_{\tau} |R_{xy}(\tau)|$$

where:

- $R_{xy}(\tau)$ is the cross-correlation between microphone signals
- τ represents the time delay
- τ^* is the estimated Time Difference of Arrival (TDOA)

Several researchers have demonstrated that TDOA provides reliable directional information when microphone synchronization and spatial geometry are properly designed. However, TDOA estimation becomes challenging in highly reverberant and noisy environments.

1.3 Microphone Array-Based Localization

Microphone arrays improve spatial sensitivity and localization accuracy by capturing acoustic signals from multiple spatially distributed sensors. Different microphone geometries including linear arrays, circular arrays, and X-configurations have been investigated in literature. Linear arrays are simple to implement but suffer from front-back ambiguity. Circular arrays provide better angular coverage but require more microphones and increased computational complexity. X-configuration arrays are commonly used in quadcopter-based systems because they naturally align with UAV arm structures and provide symmetric directional coverage. The spacing between microphones significantly affects localization performance. Larger spacing increases measurable delay resolution but may introduce phase ambiguity and synchronization challenges.

1.4 Machine Learning for Acoustic Classification

Machine learning methods have significantly improved the performance of acoustic event detection systems. Traditional classifiers such as Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), Decision Trees, and Random Forests have been widely used

for environmental sound classification. Random Forest classifiers are particularly suitable for acoustic localization tasks because they:

- Handle nonlinear relationships effectively
- Are robust against noise and overfitting
- Support multi-class classification
- Require relatively low computational resources
- Provide fast inference speed

Recent deep learning approaches use Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and transformer-based architectures for acoustic event detection. However, these approaches often require large datasets and high computational resources.

1.5 VGGish-Based Audio Feature Extraction

VGGish is a deep audio embedding model developed using a modified VGG architecture trained on large-scale audio datasets. The model converts audio waveforms into compact high-level embeddings representing important acoustic patterns. VGGish embeddings have shown strong performance in environmental sound classification tasks due to their ability to capture robust spectral and temporal representations. Compared to hand-crafted features such as MFCCs, VGGish embeddings provide improved generalization capability under noisy environmental conditions. Several studies have successfully applied VGGish embeddings to audio event detection, urban sound recognition, and surveillance applications.

1.6 Research Gap

Although substantial research has been conducted in acoustic gunshot detection and localization, several limitations remain in existing systems:

- Limited mobility due to fixed sensor installations
- Reduced performance under environmental noise
- High deployment cost of large acoustic sensor networks
- Lack of lightweight real-time onboard processing systems

- Limited integration of UAV platforms with acoustic localization frameworks

Furthermore, relatively few studies have explored real-time multi-microphone gunshot localization using UAV-mounted microphone arrays combined with machine learning techniques. This work addresses these limitations by proposing a lightweight UAV-based acoustic localization framework using TDOA feature extraction and Random Forest-based direction estimation.

Chapter 2

Methodology

The proposed system consists of two major subsystems:

1. Gunshot Detection
2. Direction of Arrival (DOA) Estimation

A quadcopter UAV equipped with a four-microphone acoustic array is used to capture environmental audio signals. The recorded multi-channel audio is processed onboard using signal processing and machine learning techniques. The overall workflow of the proposed system is illustrated in Fig. 2.1.

2.0.1 Overall System Architecture

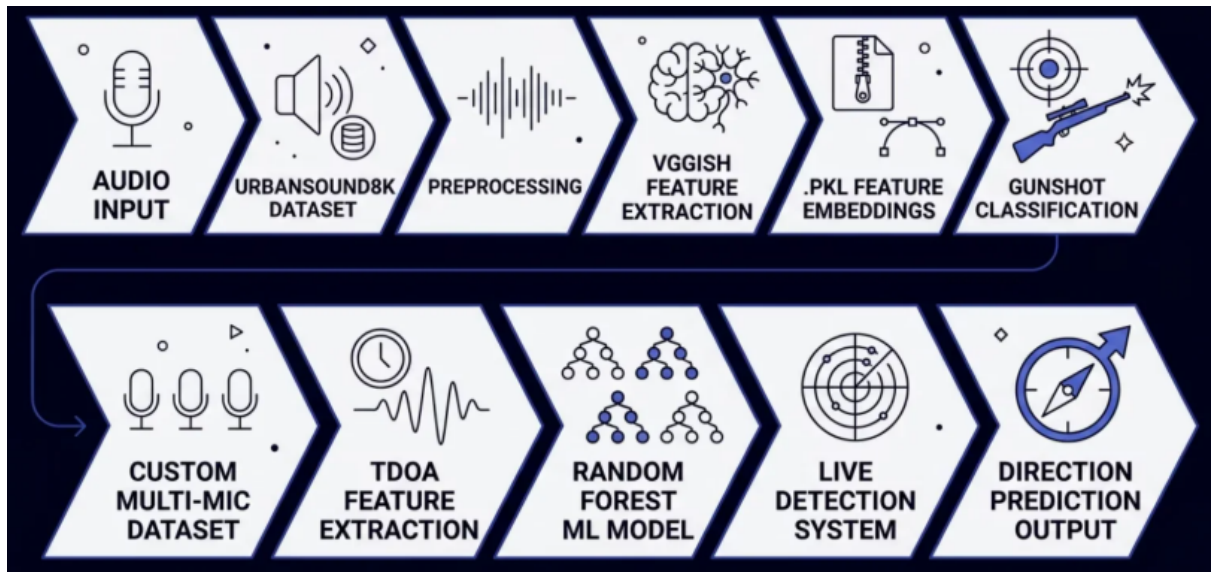


Figure 2.1: Overall architecture of the proposed UAV-based gunshot detection and localization system

The complete processing pipeline consists of the following stages:

1. Multi-channel acoustic acquisition
2. Signal preprocessing
3. Feature extraction
4. Gunshot classification
5. Time Difference of Arrival (TDOA) estimation
6. Direction classification
7. Real-time output generation

2.0.2 Drone Platform

The hardware platform used in this work is the Holybro X500 V2 quadcopter frame integrated with onboard processing and acoustic sensing modules.

The major hardware components include:

- Pixhawk flight controller
- NVIDIA Jetson Nano
- GPS module
- Four omnidirectional microphones
- Power distribution module

The Jetson Nano performs real-time audio processing and machine learning inference, while the Pixhawk controller maintains stable UAV flight operations.

2.0.3 Microphone Array Geometry

A four-microphone array arranged in an X-shaped configuration was designed for spatial acoustic sensing.

The microphones were mounted at the arm tips of the UAV to maximize the baseline distance between sensors.

The placement configuration provides:

- Uniform 360° directional coverage

- Improved TDOA sensitivity
- Better angular discrimination
- Reduced localization ambiguity

The geometry of the microphone array is shown in Fig. 2.2.



Figure 2.2: Four-microphone array geometry mounted on the UAV platform

2.0.4 Audio Acquisition

The microphones simultaneously capture impulsive acoustic events from different spatial directions. The recordings were collected using the following parameters:

Table 2.1: Audio Acquisition Parameters

Parameter	Value
Sampling Rate	48 kHz
Bit Depth	16-bit
Channels	4
Recording Format	WAV
Clip Duration	3 seconds

A high sampling rate was selected to preserve transient gunshot characteristics and improve TDOA estimation accuracy.

2.0.5 Signal Preprocessing

The acquired audio signals were preprocessed before feature extraction.

The preprocessing pipeline included:

- DC offset removal
- Signal normalization
- Noise suppression
- Windowing
- Bandpass filtering

Bandpass filtering was applied to suppress low-frequency drone propeller noise and high-frequency environmental interference. The normalized signal is represented as:

$$x_n = \frac{x - \mu}{\sigma}$$

where:

- x is the original signal
- μ is the mean
- σ is the standard deviation

2.0.6 Feature Extraction

Both time-domain and frequency-domain features were extracted from the acoustic recordings. The extracted features include:

- Spectral Centroid
- Spectral Rolloff
- Frequency Band Energy
- Intensity Ratios
- Onset Time Differences
- Cross-correlation delays

These features help distinguish gunshots from other impulsive environmental sounds.

2.0.7 Cross-Correlation-Based TDOA Estimation

The Time Difference of Arrival (TDOA) between microphone pairs was estimated using cross-correlation analysis. The cross-correlation function is defined as: The delay corresponding to the maximum correlation peak is computed as: The estimated delays are used to infer the direction of the sound source relative to the UAV.

2.0.8 Direction Estimation

Direction estimation was performed using relative delay information between microphones.

The direction angle is computed using:

where:

- L_{LR} represents left-right delay
- L_{FB} represents front-back delay

The angular space was divided into 12 directional classes with 30° separation.

2.0.9 Machine Learning Models

Two machine learning models were used in this work.

Gunshot Detection Model

Gunshot detection was performed using VGGish embeddings and a deep neural network classifier. The VGGish model extracts compact high-level audio embeddings that effectively represent acoustic patterns. Advantages of using VGGish include:

- Robust feature learning
- Noise resilience
- Improved classification accuracy
- Compact embedding generation

Direction Classification Model

Direction classification was performed using a Random Forest classifier.

Random Forest was selected because of:

- Strong multi-class classification capability
- Robustness against overfitting
- Fast inference speed
- Ability to handle nonlinear relationships

The classifier was trained using extracted TDOA and spectral features.

2.0.10 Real-Time Processing

The proposed system was designed for real-time onboard inference.

The processing pipeline executes the following tasks sequentially:

1. Continuous audio acquisition
2. Gunshot event detection
3. Feature extraction
4. Direction estimation
5. Output generation

The Jetson Nano performs inference locally without requiring cloud communication, enabling deployment in remote environments.

Chapter 3

Experimental Setup and Dataset Collection

This section describes the experimental environment, dataset preparation methodology, recording procedure, and hardware configuration used for evaluating the proposed UAV-based gunshot detection and direction estimation system.

3.0.1 Experimental Environment

The experiments were conducted in an open outdoor environment to minimize excessive acoustic reflections and reverberation effects.

The testing area was selected to ensure:

- Minimal obstruction
- Reduced environmental echo
- Controlled background noise
- Safe UAV operation

The UAV remained stationary during audio acquisition to reduce motion-induced noise and maintain stable microphone geometry.

3.0.2 Hardware Configuration

The proposed system consists of the following hardware components:

Table 3.1: Hardware Components Used in the Proposed System

Component	Description
UAV Platform	Holybro X500 V2
Flight Controller	Pixhawk
Processing Unit	NVIDIA Jetson Nano
Microphones	4 Omnidirectional Microphones
Audio Interface	USB Multi-channel Interface

The microphones were mounted symmetrically on the drone arms to maximize spatial separation and improve localization accuracy.

3.0.3 Directional Data Collection

A custom multi-channel directional dataset was collected for training and evaluating the direction estimation model.

The data collection setup consisted of:

- 12 directional positions
- 30° angular spacing
- Constant radial distance of 11 meters
- Four-channel synchronized audio recording

The directional classes are represented as:

$$0^\circ, 30^\circ, 60^\circ, \dots, 330^\circ$$

The experimental setup for directional recording is illustrated in Fig. 3.1.

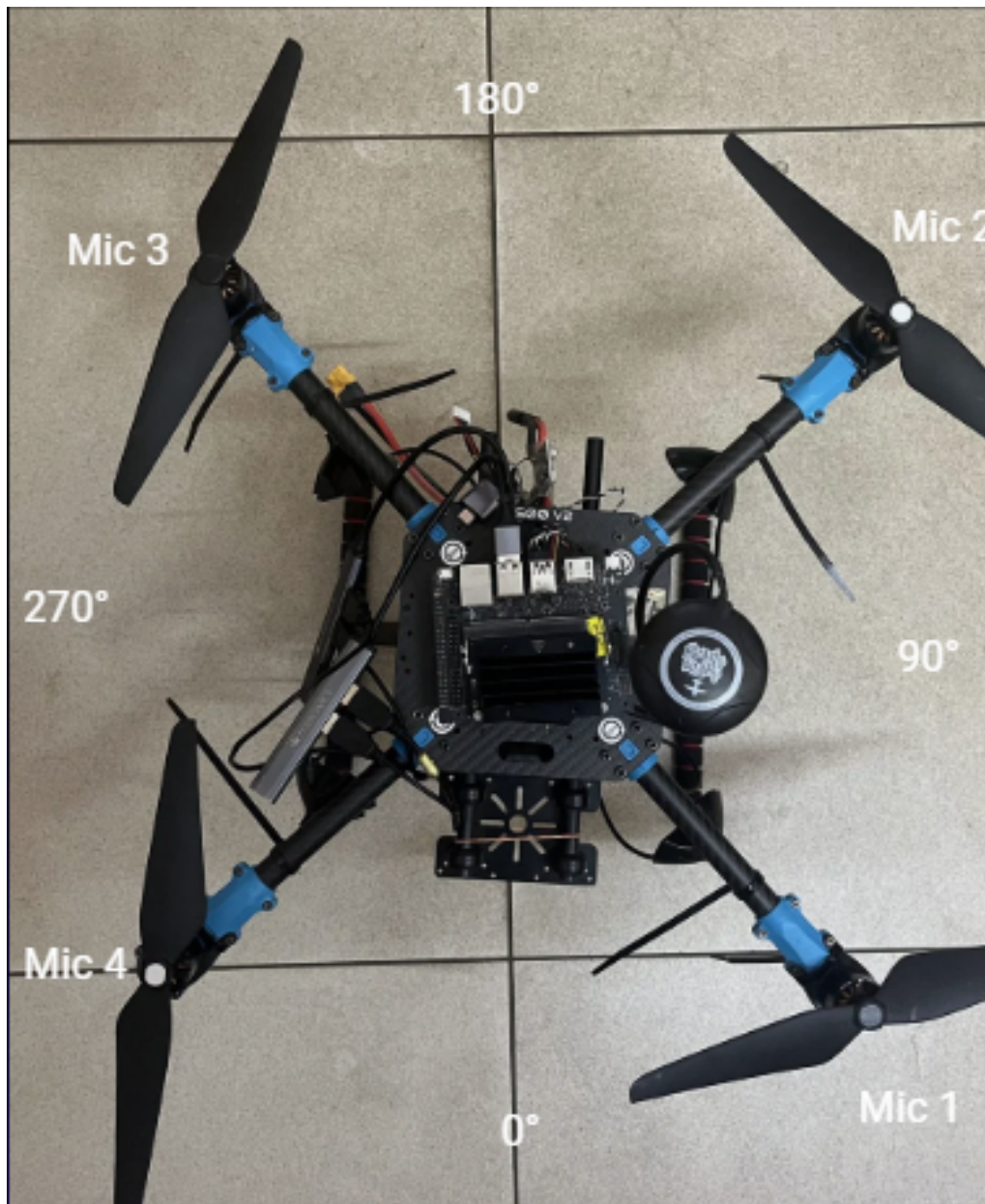


Figure 3.1: Directional dataset collection setup

3.0.4 Gunshot Audio Collection

Due to safety and legal constraints associated with real firearm usage, controlled gunshot-like acoustic signals were generated using high-intensity impulsive sound playback systems.

The dataset included:

- Simulated gunshot sounds
- Firecracker sounds
- Tyre burst sounds

- Environmental background noise

Including additional impulsive sounds helped improve model robustness and reduce false positive detections.

3.0.5 UrbanSound8K Dataset

The UrbanSound8K dataset was used for training the gunshot detection model. The dataset contains urban environmental audio recordings belonging to multiple sound categories. The primary target class used in this work was:

- `gun_shot`

The dataset characteristics are summarized in Table 3.2.

Table 3.2: UrbanSound8K Dataset Statistics

Parameter	Value
Total Audio Clips	8732
Target Class	<code>gun_shot</code>
Sampling Rate	48 kHz
Audio Format	WAV
Classes	10
Dataset Size	5.61 GB

3.0.6 Recording Procedure

The recording procedure followed a structured acquisition process to maintain synchronization and consistency across all directional classes.

The procedure included:

1. Positioning the sound source at predefined angular locations
2. Maintaining a fixed source-to-UAV distance
3. Simultaneous multi-channel audio recording
4. Repeating recordings for multiple samples **upto(50) samples**
5. Labeling audio clips according to direction class

Each recording duration was approximately 3 seconds.

3.0.7 Dataset Labeling

The recorded audio files were manually labeled according to their corresponding directional class.

An example naming convention is shown below:

`angle_090_sample01.wav`

where:

- 090 represents the source direction
- sample01 represents the sample number

Proper labeling is important for supervised machine learning training.

3.0.8 Train-Test Split

The dataset was divided into training and testing subsets for model evaluation.

The split ratio used was:

- Training Set: 80%
- Testing Set: 20%

The split was performed randomly while maintaining balanced representation across all directional classes.

3.0.9 Challenges During Data Collection

Several practical challenges were encountered during dataset acquisition.

These included:

- Drone propeller noise
- Wind interference
- Environmental background sounds
- Synchronization accuracy
- Limited outdoor testing duration

Special care was taken to minimize environmental disturbances during recording sessions.

3.0.10 System Parameters

The final experimental parameters used in the proposed system are summarized in Table 3.3.

Table 3.3: System Parameters

Parameter	Value
Sampling Rate	48 kHz
Number of Microphones	4
Direction Classes	12
Angular Resolution	30°
Source Distance	11 m
Audio Duration	3 s
Speed of Sound	343 m/s

Chapter 4

Results and Performance Evaluation

This section presents the experimental results obtained for gunshot detection and direction estimation using the proposed UAV-based acoustic surveillance framework.

The performance of the proposed system was evaluated using classification accuracy, confusion matrices, precision, recall, and F1-score metrics.

4.0.1 Gunshot Detection Performance

The gunshot detection subsystem was evaluated using VGGish embeddings and a feed-forward neural network classifier. The model was trained to distinguish gunshot sounds from other environmental and impulsive sounds. The proposed detection model achieved an overall classification accuracy of:

$$92.47\%$$

The high classification accuracy demonstrates the effectiveness of deep audio embeddings for impulsive sound recognition.

Gunshot Detection Classification Report

The classification report for the gunshot detection model is shown in Fig. [4.1](#).

Classification Report:				
	precision	recall	f1-score	support
air_conditioner	0.85	0.97	0.91	191
car_horn	0.95	0.85	0.90	61
children_playing	0.89	0.94	0.92	201
dog_bark	0.96	0.86	0.91	192
drilling	0.94	0.93	0.94	178
engine_idling	0.98	0.89	0.93	203
gun_shot	1.00	1.00	1.00	63
jackhammer	0.93	0.95	0.94	209
siren	0.90	0.95	0.92	170
street_music	0.93	0.90	0.91	192
accuracy			0.92	1660
macro avg	0.93	0.93	0.93	1660
weighted avg	0.93	0.92	0.92	1660
Final Test Accuracy: 0.9247				

Figure 4.1: Classification report and accuracy for gunshot detection

The classification metrics indicate strong precision and recall values across the target class.

Gunshot Detection Confusion Matrix

The confusion matrix for gunshot detection is presented in Fig. [4.2](#).

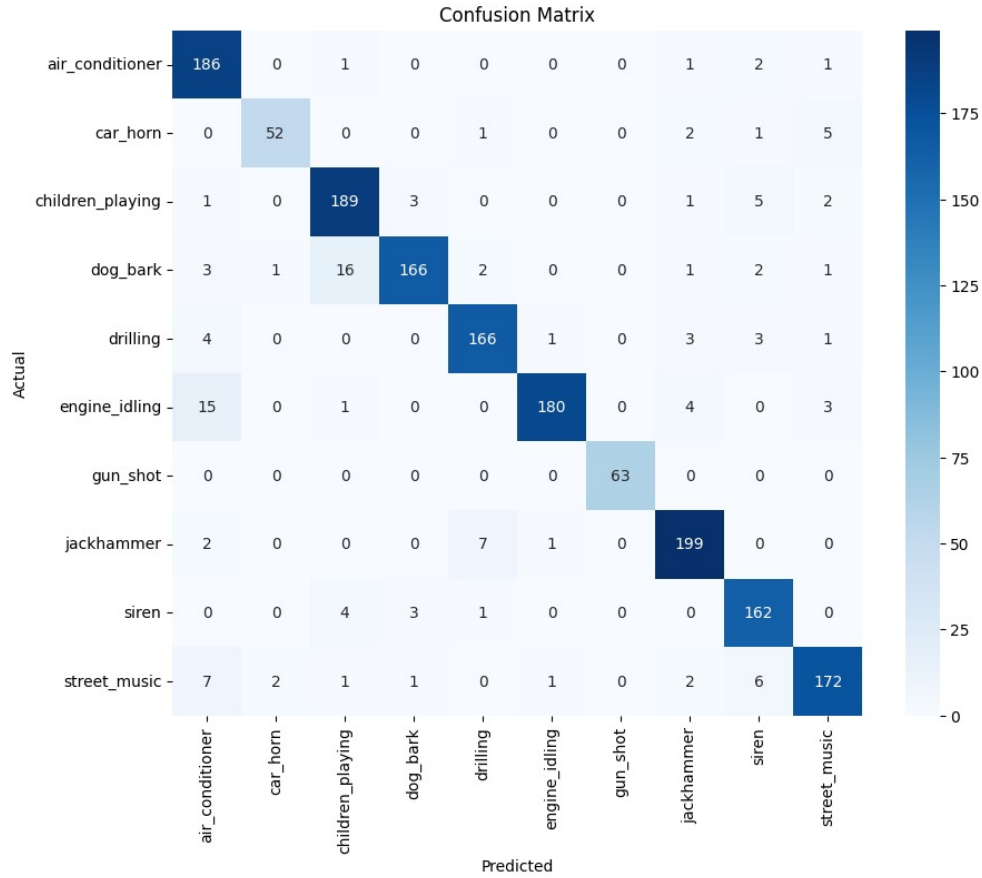


Figure 4.2: Confusion matrix for gunshot detection

The confusion matrix shows that the model successfully differentiates gunshots from other environmental sounds with very low false positive rates.

4.0.2 Direction Estimation Performance

Direction estimation was performed using extracted TDOA and spectral features with a Random Forest classifier. The angular space was divided into 12 directional classes separated by 30° intervals. The proposed direction estimation model achieved an overall accuracy of:

$$84.6\%$$

The results demonstrate that the proposed microphone array geometry and feature extraction framework can effectively estimate source direction in real-world outdoor conditions.

Direction Classification Results

The classification report for direction estimation is shown in Fig. 4.3.


```

Classes: ['0°' '120°' '150°' '180°' '210°' '240°' '270°' '300°' '30°' '330°' '60°'
'90°']
Samples per class: [47 45 49 50 50 50 50 49 42 50 50 50]

Training Random Forest...
Cross-validation accuracy: 0.847 (+/- 0.042)

Test set accuracy: 0.846

Classification Report:

```

	precision	recall	f1-score	support
0°	1.00	1.00	1.00	10
120°	0.62	0.56	0.59	9
150°	0.71	0.50	0.59	10
180°	0.78	0.70	0.74	10
210°	0.75	0.90	0.82	10
240°	1.00	0.90	0.95	10
270°	1.00	1.00	1.00	10
300°	1.00	0.90	0.95	10
30°	0.80	1.00	0.89	8
330°	0.83	1.00	0.91	10
60°	0.90	0.90	0.90	10
90°	0.73	0.80	0.76	10
accuracy			0.85	117
macro avg	0.84	0.85	0.84	117
weighted avg	0.85	0.85	0.84	117

Figure 4.3: Direction estimation classification report and accuracy

The results indicate that the classifier performs strongly for most directional classes with high precision and recall values.

Direction Estimation Confusion Matrix

The confusion matrix for direction estimation is shown in Fig. 4.4.

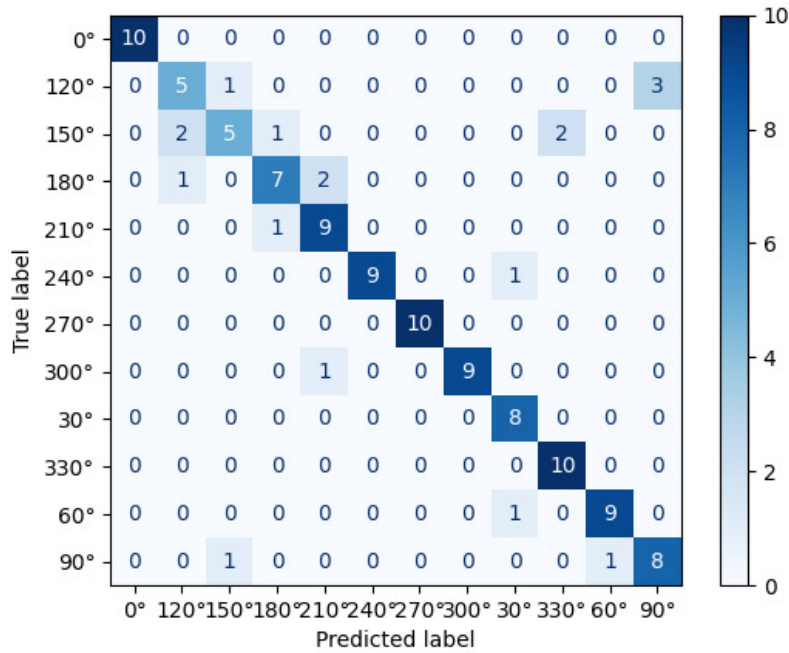


Figure 4.4: Confusion matrix for direction estimation

The confusion matrix demonstrates that most prediction errors occur between neighboring angular classes because adjacent directions produce similar TDOA patterns.

4.0.3 Feature Importance Analysis

The Random Forest classifier provides feature importance estimation for understanding the contribution of individual features.

The most important features observed during training were:

- TDOA delays
- Intensity ratios
- Onset timing differences
- Spectral centroid
- RMS energy

These features contributed significantly toward distinguishing directional classes.

4.0.4 Real-Time Processing Performance

The proposed framework was tested for real-time onboard inference using the NVIDIA Jetson Nano.

The system successfully performed:

- Continuous audio acquisition
- Real-time preprocessing
- Gunshot classification
- Direction estimation

The average inference latency was sufficiently low for practical real-time surveillance applications.

4.0.5 Comparative Discussion

Compared with traditional fixed surveillance systems, the proposed UAV-based framework offers several advantages:

- Mobile deployment capability
- Large-area coverage
- Reduced infrastructure dependency
- Improved accessibility in difficult terrains
- Better operation in low-visibility environments

The combination of acoustic sensing and machine learning enables efficient impulsive sound analysis under dynamic environmental conditions.

4.0.6 Limitations of Experimental Results

Although promising performance was achieved, several limitations remain.

These include:

- Simulated gunshot audio instead of real firearm discharge
- Limited dataset size
- Fixed-distance recordings

- Environmental noise sensitivity
- Limited angular resolution

Future improvements can address these limitations using larger datasets and advanced noise suppression methods.

4.1 Discussion

The experimental results demonstrate the feasibility of using UAV-mounted acoustic sensing systems for real-time gunshot detection and directional localization. The proposed framework combines signal processing and machine learning techniques to achieve reliable impulsive sound analysis under outdoor conditions. The use of a four-microphone array improved spatial sensing capability and enabled effective estimation of Time Difference of Arrival (TDOA) features. The VGGish-based gunshot detection subsystem demonstrated strong classification performance due to its ability to learn high-level acoustic representations from environmental audio data. Similarly, the Random Forest classifier provided reliable directional prediction because of its robustness against nonlinear feature distributions and environmental noise. The obtained results indicate that acoustic sensing can complement traditional optical surveillance systems, especially in situations where visibility is poor due to darkness, smoke, fog, or physical obstruction.

4.1.1 Importance of Acoustic Surveillance

Acoustic surveillance systems provide several advantages compared to conventional camera-based systems. These include:

- Operation in low-light conditions
- Detection beyond visual line-of-sight
- Lower dependency on environmental illumination
- Capability to identify impulsive threat events

The mobility of UAV platforms further enhances surveillance flexibility and allows rapid deployment across large geographical regions.

4.1.2 Effect of Drone Noise

One of the major challenges associated with UAV-based acoustic sensing is propeller noise generated by the drone motors.

Drone noise introduces:

- Low-frequency interference
- Spectral distortion
- Reduced signal-to-noise ratio
- Errors in TDOA estimation

To minimize these effects, microphones were mounted at the arm tips and preprocessing filters were applied to suppress low-frequency components.

Future systems may incorporate adaptive noise cancellation and beamforming methods for improved robustness.

4.1.3 Importance of Microphone Geometry

The geometry of the microphone array plays a critical role in localization accuracy.

The X-shaped four-microphone arrangement provided:

- Improved angular discrimination
- Better directional sensitivity
- Increased spatial coverage
- Reduced localization ambiguity

Increasing the distance between microphones improved measurable delay resolution and enhanced directional estimation performance.

4.2 Applications

The proposed UAV-based acoustic surveillance framework has potential applications across multiple domains.

4.2.1 Smart Surveillance

The system can be integrated into intelligent surveillance networks for detecting firearm activity in urban environments.

Potential applications include:

- Public safety monitoring
- Crime detection
- Emergency response systems

4.2.2 Military and Defense

In military applications, the proposed framework can assist in:

- Battlefield monitoring
- Sniper localization
- Border surveillance
- Threat detection

The UAV platform enables deployment in hazardous or inaccessible regions without exposing personnel to direct danger.

4.2.3 Wildlife Protection

Illegal poaching activities often involve firearm usage in protected forest regions.

The proposed system can assist wildlife protection agencies by:

- Detecting gunshot events
- Monitoring remote forest regions
- Supporting anti-poaching operations

4.2.4 Disaster and Emergency Response

The system may also assist emergency responders in identifying threat zones during disaster or conflict situations.

Real-time localization capability can improve situational awareness and response coordination.

Chapter 5

Future Scope

Although promising performance was achieved, several improvements can be implemented in future work.

5.0.1 Real Firearm Testing

Future experiments should include recordings from actual firearm discharges to capture realistic acoustic shockwave characteristics and reverberation effects. This would improve model generalization and practical deployment reliability.

5.0.2 Higher Directional Resolution

The current system uses 30° directional intervals.

Future systems can improve localization precision using:

- Additional microphones
- Higher sampling rates
- Advanced beamforming algorithms

5.0.3 Adaptive Noise Cancellation

Advanced noise suppression methods can improve robustness against:

- Drone propeller noise
- Wind interference
- Urban environmental noise

Machine learning-based denoising and adaptive filtering techniques may significantly improve performance.

5.0.4 Autonomous UAV Navigation

Future systems may integrate autonomous navigation algorithms that enable the UAV to automatically move toward the detected gunshot source.

Such systems could support:

- Dynamic threat tracking
- Autonomous surveillance
- Real-time target monitoring

5.0.5 GPS-Based Source Localization

Future work may combine directional estimation with GPS data from multiple UAVs for accurate triangulation of gunshot source location. Multi-UAV cooperative sensing can significantly improve localization accuracy across large environments.

5.1 Conclusion

This paper presented a UAV-based acoustic surveillance framework for gunshot detection and direction estimation using signal processing and machine learning techniques.

The proposed system integrates:

- A four-microphone acoustic array
- Cross-correlation-based TDOA estimation
- VGGish-based gunshot detection
- Random Forest-based direction classification

Experimental evaluation demonstrated:

- 92.47% gunshot detection accuracy
- 84.6% direction estimation accuracy

The obtained results confirm the feasibility of real-time UAV-based acoustic surveillance for security, defense, and emergency response applications. The proposed framework demonstrates that combining UAV mobility with machine learning-based acoustic sensing can provide an efficient alternative to conventional fixed surveillance systems. Future improvements involving real firearm datasets, adaptive noise suppression, and multi-UAV cooperation can further enhance localization accuracy and deployment robustness.

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